

1. unionSets(ArrayList<Integer> listA, ArrayList<Integer> listB)

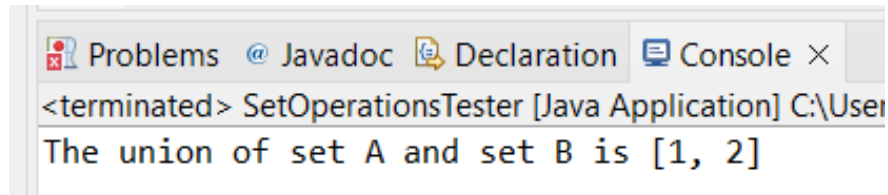
a. Test Case 1: the same set: {1, 2}

i. Expected Output: {1, 2}

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1,2));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(1,2));
9
10    x.unionSets(listA, listB);
11
12 }
```

iii. Output:



The screenshot shows the IDE's console window with the following text:
<terminated> SetOperationsTester [Java Application] C:\User
The union of set A and set B is [1, 2]

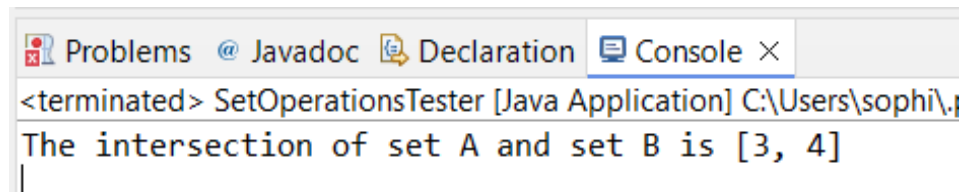
b. Test Case 2: overlapping sets: {1, 2, 3, 4}, {3, 4, 5, 6}

i. Expected Output: {1,2, 3, 4, 5, 6}

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1,2, 3, 4));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));
9
10    x.intersectSets(listA, listB);
11
12 }
```

iii. Output:



The screenshot shows the IDE's console window with the following text:
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.
The intersection of set A and set B is [3, 4]

2. intersectSets(ArrayList<Integer> listA, ArrayList<Integer> listB)

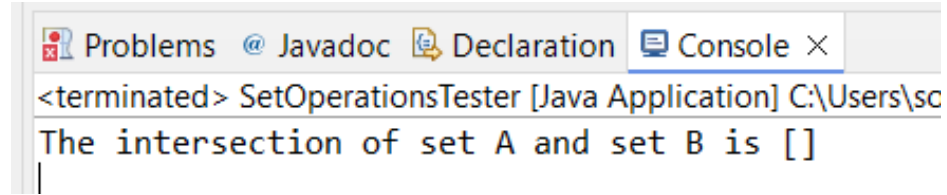
a. Test Case 1: no overlap

i. Expected Output: {}

ii. Input:

```
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1,2));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4));
9
10    x.intersectSets(listA, listB);
11 }
```

iii. Output:



b. Test Case 2: overlapping sets: {1, 2, 3, 4}, {3, 4, 5, 6}

i. Expected Output: {3, 4}

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));
9
10    x.intersectSets(listA, listB);
11
12 }
```

iii. Output:

3. findComplement(ArrayList<Integer> universalList, ArrayList<Integer> listA)

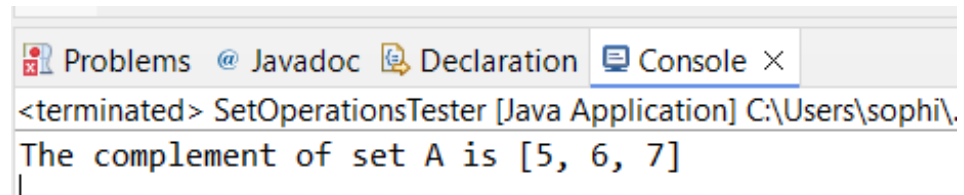
a. Test Case 1: setA: {1, 2, 3, 4}, Universal set: {1, 2, 3, 4, 5, 6, 7}

i. Expected Output: {5, 6, 7}

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1, 2, 3, 4, 5, 6, 7));
10    x.findComplement(universalList, listA);
11
12 }
```

iii. Output:



4. findDifference(ArrayList<Integer> listA, ArrayList<Integer> listB)

a. Test Case 1: setA: {1, 2, 3, 4}, Universal set: {3, 4, 5, 6}

i. Expected Output: A – B: {1, 2}, B – A: {5, 6}

ii. Input:

```
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1, 2, 3, 4, 5, 6, 7));
10    x.findDifference(listA, listB);
11 }
```

iii. Output:

5. findSymmetricDifference(ArrayList<Integer> listA, ArrayList<Integer> listB)

a. Test Case 1: Set A: {1, 2, 3, 4}, Set B = {3, 4, 5, 6}

i. Expected Output: {1, 2, 5, 6}

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    x.findSymmetricDifference(listA, listB);
11 }
```

iii. Output:

6. findCrossProduct(ArrayList<Integer> listA, ArrayList<Integer> listB)

a. Test Case 1: Set A: {1, 2}, Set B = {3, 4}

i. Expected Output: {(1, 3), (1, 4), (2, 3), (2, 4)}

ii. Input:

```
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    x.findCrossProduct(listA, listB);
11 }
```

iii. Output:

7. getPowerSet(ArrayList<Integer> listA)

a. Test Case 2: (1, 2, 3)

i. Expected Output: {}, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}

ii. Input:

```

5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    x.getPowerSet(listA);
11
12 }

```

iii. Output:

```

Problems @ Javadoc Declaration Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_23.0
The power set of set A is [[], [1], [2], [1, 2], [3], [1, 3], [2, 3], [1, 2, 3]]

```

8. `checkIfSubset(ArrayList<Integer> possibleSubset, ArrayList<Integer> possibleSuperset)`

a. Test Case 1: Is not a subset

i. Expected Not a subset

ii. Input:

```

4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    x.checkIfSubset(listA, listB);
11

```

iii. Output:

```

Problems @ Javadoc Declaration Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\p2\pool\pl
The first list is not a subset of the second list

```

b. Test Case 2: Is a subset

i. Expected Is a subset

ii. Input:

```

4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    x.checkIfSubset(listA, listB);
11

```

iii. Output:

```

Problems @ Javadoc Declaration Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\p2
The first list is a subset of the second list

```

9. `checkIfProperSubset(ArrayList<Integer> possibleSubset, ArrayList<Integer> possibleSuperset)`

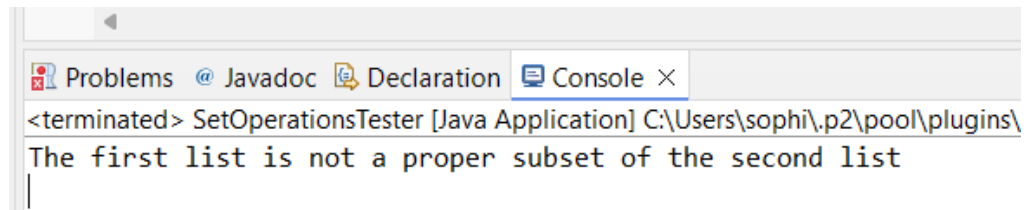
a. Test Case 1: Is a proper subset (same sets)

i. Expected Not a proper subset

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(1, 2, 3));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    x.checkIfProperSubset(listA, listB);
11
12 }
```

iii. Output:



The screenshot shows the IDE's console window with the following text: `<terminated> SetOperationsTester [Java Application] C:\Users\sophi\p2\pool\plugins\` followed by the message `The first list is not a proper subset of the second list`.

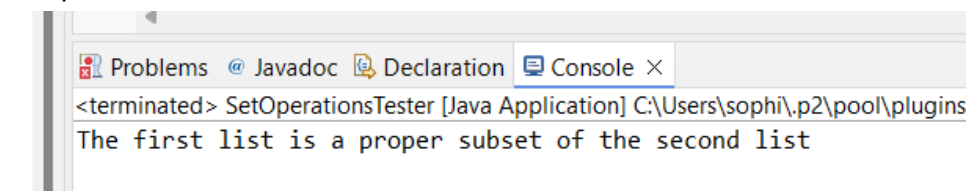
b. Test Case 2: Is a proper subset

i. Expected Is a proper subset

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(1, 2, 3, 4, 5));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    x.checkIfProperSubset(listA, listB);
11
12 }
```

iii. Output:



The screenshot shows the IDE's console window with the following text: `<terminated> SetOperationsTester [Java Application] C:\Users\sophi\p2\pool\plugins\` followed by the message `The first list is a proper subset of the second list`.

10. checkIfDisjoint(ArrayList<Integer> listA, ArrayList<Integer> listB)

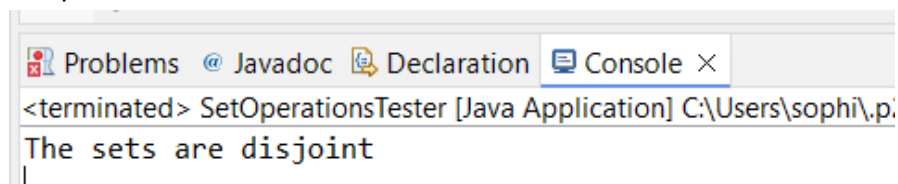
a. Test Case 1: are disjoint

i. Expected Sets are disjoint

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations x = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(4, 5, 6));
9     ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    x.checkIfDisjoint(listA, listB);
11
12 }
```

iii. Output:



The screenshot shows the IDE's console window with the following text: `<terminated> SetOperationsTester [Java Application] C:\Users\sophi\p2\pool\plugins\` followed by the message `The sets are disjoint`.

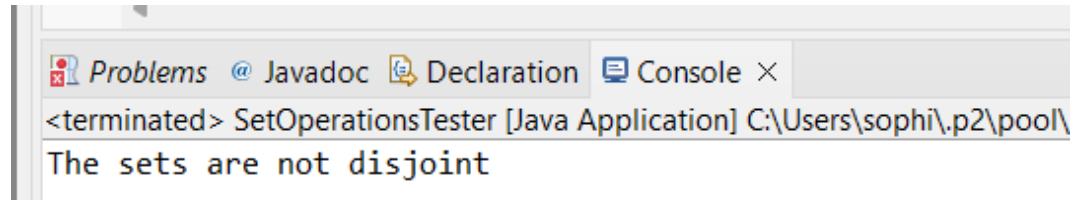
b. Test Case 2: Are not disjoint

i. Expected Subsets are not disjoint

ii. Input:

```
4
5 public static void main(String[] args) {
6     Operations o = new Operations();
7     ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
8     ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(4, 5, 6 ));
9     ArrayList<Integer> universallist = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
10    System.out.println("The sets are not disjoint\n\n");
11    o.checkIfDisjoint(listA, listB);
12 }
```

iii. Output:



The screenshot shows an IDE console window with tabs for Problems, Javadoc, Declaration, and Console. The Console tab is active, displaying the output of a Java application. The output text is: <terminated> SetOperationsTester [Java Application] C:\Users\sophi\p2\pool\ The sets are not disjoint